WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-10-12 to 2026-10-17

Curated articles in political communication and computational social science.

SPECIAL ISSUE UPDATES

Calls checked: 2026-10-17.

Manual calls omitted pending renewed official-page verification: Public Relations Review.

Special Issue: How Communication Technology Mediates Death, Dying, and Grief
Western Journal of Communication | Deadline within 14 days | call for papers
Deadline: 2026-10-31
Official call / submission details

Special Collection: The new intelligence age: surveillance, espionage and power
Humanities & Social Sciences Communications | Newly found | call for papers
Deadline: 2027-06-09
Official call / submission details

Thematic Issue: Transformation of Public Communication in the Era of AI: Between Disruption, Adaptation, and Reinvention
Media and Communication | Newly found | call for papers
Deadline: 2027-09-15
Official call / submission details

SUMMARY

This week, COMAP Journal Bot examines how AI changes political behavior-and how researchers can tell whether AI-generated evidence resembles human evidence. New studies test consequential political persuasion, detect automated survey respondents, benchmark synthetic preferences across 74 countries, and validate generated social-media language. A political-methods contribution offers a targeted approach to sparse text-classification categories.

The institutional reading connects authoritarian demolition, declining public services and populist support, academic freedom, and the durability of inequality. Work on information-driven network change, online vaccine debate, smartphone policies, policy evasion and urban vulnerability broadens the picture; catch-up papers add visual persuasion and the local credibility of AI-assisted election verification.

You may have received two September 7 emails because the first was sent before that collection window closed; the second completed the same edition. Saturday delivery now aligns collection and sending, so each future email represents a distinct Monday-Saturday edition.

HIGHLIGHTS

AI can persuade people to take political actions

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-10-16

Why it matters · Two preregistered experiments test actual petition signing and donations, showing why attitude changes alone can misstate AI's behavioral influence.

Open article →

Estimating the threat of AI-agent responding across online survey platforms

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-10-15

Why it matters · Validates detection tests across seven survey platforms and shows how undetected AI agents can distort social-science results.

Open article →

Simulated responses from large language models are weak proxies for human economic preferences across 74 countries

NATURE HUMAN BEHAVIOUR · 2026-10-12

Why it matters · Benchmarks simulated economic preferences against 67,201 respondents in 74 countries, exposing limits in cross-national representativeness.

Open article →

AI-generated language remains systematically distinguishable from human communication

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-10-12

Why it matters · Introduces CHAT to evaluate generated social-media language, identifying trade-offs between human likeness and fidelity to human content.

Open article →

The Synthetic Imputation Approach: Using LLMs to Generate Synthetic Texts for Underrepresented Categories in Supervised Classification Tasks

POLITICAL ANALYSIS · 2026-10-14

Why it matters · Targets sparse annotation categories with synthetic texts and tests when augmentation improves political text classification.

Open article →

Spatiality of preemptive coercion: How autocrats demolish informal settlements for political control

AMERICAN JOURNAL OF POLITICAL SCIENCE · 2026-10-14

Why it matters · Combines satellite imagery, digital street views and fieldwork to examine demolition as preemptive authoritarian control.

Open article →

Educational intervention can rewire village social networks

NATURE · 2026-10-14

Why it matters · A village-level randomized intervention changes friendship, financial and advice ties, treating information provision as a force that reorganizes social structure.

Open article →

Two Paths of Conversation: How Stories and Statistics are Associated With Community Engagement in Online Debate Surrounding Vaccine Science Skeptical Content

COMMUNICATION RESEARCH · 2026-10-17

Why it matters · Uses LLM-assisted analysis of Reddit vaccine debate to distinguish the participation patterns associated with statistical and narrative evidence.

Open article →

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- Science
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Previous Week Catch-Up

- Communication Research
- Nature Communications
- Political Communication

COLLECTION SNAPSHOT

TOTAL COLLECTED CANDIDATES	NEW THIS WEEK CANDIDATES	PREVIOUS WEEK CATCH-UP CANDIDATES
348	150	197
LATE ADDITIONS CANDIDATES		
1		

Curated digest below: 23 new, 3 catch-up, 0 late additions

These candidates passed automated source filters; counts precede manual relevance filtering. The full curated digest and attached PDF apply the stricter COMAP threshold.

All 17 article sources completed for September 20-October 17, 2026, with no source failures or recorded optional-enrichment errors. October 10 selections are not repeated.

FULL CURATED DIGEST

New This Week

American Journal of Political Science

Spatiality of preemptive coercion: How autocrats demolish informal settlements for political control

2026-10-14 · DOI 10.1111/AJPS.70085

Shiqi Ma

Department of Politics and International Studies SOAS University of London London United Kingdom

Abstract. Abstract Why do authorities demolish informal settlements in metropolises? The social sciences have a long tradition of addressing this issue through the lens of economic development or welfare provision. I instead propose a theory of political control. Autocrats adopt demolition as a preemptive and coercive tool to remove communities substantively and symbolically less legible to them, thereby ensuring societal compliance and restoring visual order. Empirically, I compile a unique dataset of 320 demolished and undemolished informal settlements in Beijing between 2017 and 2018, based on satellite imagery, digital street views, and fieldwork. The results indicate that the low legibility of informal settlements to authorities correlates with a greater likelihood of demolition. This paper contributes to the social science literature by connecting legibility politics with urban governance. It also introduces a spatial perspective and leverages novel data sources to deepen our understanding of preemptive coercion under authoritarianism.

Read article →

American Political Science Review

Public Service Decline and Support for the Populist Right: Evidence from England's National Health Service

2026-10-13 · DOI 10.1017/S0003055426101944

ZACHARY P. DICKSON; SARA B. HOBOLT; CATHERINE DE VRIES; SIMONE CREMASCHI

IE University; London School of Economics and Political Science; University of Oxford

Abstract. The rise of populist right parties is well studied, but relatively little attention has been given to how public service performance influences voter support. Given that public services are often the primary means through which citizens interact with the state, we argue that declining public services can create grievances that increase the appeal of populist right parties. Focusing on England's National Health Service, we combine administrative data on local health facility closures with panel data on public preferences and voting intentions. Using a staggered difference-in-differences design, we find that closures reduce satisfaction with public services and increase support for populist right parties. These effects are moderated by migrant registrations at local health practices, highlighting the interaction between public service performance and immigration concerns in fueling populist right support. Our findings underscore the role of public service decline as a driver of support for populist parties, especially in areas undergoing demographic change.

Read article →

Communication Research

Two Paths of Conversation: How Stories and Statistics are Associated With Community Engagement in Online Debate Surrounding Vaccine Science Skeptical Content

2026-10-17 · DOI 10.1177/00936502261496144

Han Li; Inhwan Bae; Pengfei Zhao; Natalie Bazarova; Drew Margolin

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Abstract. Statistical and narrative evidence represent two distinct ways of substantiating claims in public reasoning, but does this distinction matter for how people engage with vaccine science skeptical (VSS) content and the discursive spaces surrounding it? This study addresses this question through a large-scale, LLM-assisted observational analysis of online engagement with VSS posts and comments that support or counter such content using statistical versus narrative evidence in the Reddit community r/DebateVaccines . Building on Fisher's distinction between rational and narrative paradigms , we theorize two paths of conversation through which evidentiary forms are associated with audience engagement with contentious vaccine discourses in a naturalistic social media setting: a rational pathway, marked by concentrated participation and corrective contestation around statistical claims and counterclaims, tending toward a single evidence-based truth "approved" by one-click reactions; and a narrative pathway, characterized by generative engagement featuring broader and more distributed participation, more discursive and generative engagement, and greater acceptance of competing viewpoints. These findings provide a springboard for theorizing evidentiary form as a key epistemic logic associated with how conversations evolve in real-world engagement with contentious, socially consequential issues such as vaccine misinformation, while informing hybrid approaches to addressing skepticism toward established vaccine science.

Read article →

Journal of Communication

Academic freedom: defending a public good

2026-10-16 · DOI 10.1093/JOC/JQAG017

Thomas Hanitzsch

Department of Media and Communication, LMU Munich , Munich,

Abstract. Abstract This essay, an expanded version of a Presidential Address delivered at the 2026 conference of the International Communication Association (ICA), addresses the global erosion of academic freedom and its implications for communication scholarship. Drawing on a survey of 546 ICA members, the article documents substantial restrictions on academic freedom, alongside widespread occupational intimidation. Most survey respondents have personally experienced hostility or intimidation related to their work, most commonly through social media and personal insults. Nearly half of them report altering their academic practice-e.g., by adjusting their teaching content and research topics or speaking out less publicly. While most attacks originate from the public and political actors, the survey shows that a significant amount of intimidation originates from colleagues and other academics. The essay argues that these findings demand a coordinated collective response, including institutional advocacy, solidarity networks, and recognition of academic freedom as an indispensable human right.

Read article →

Journal of Computer-Mediated Communication

The link between three smartphone school policies and students' mental health: a growth trajectory approach

2026-10-14 · DOI 10.1093/JCMC/ZMAG025

Lara Schreurs; Robyn Vanherle; Jana Vereecke; Lise-Marie Nassen; Laura Vandenbosch; Kathleen Beullens

Amsterdam School of Communication Research (ASCoR), University of Amsterdam , Amsterdam,; Faculty of Social Sciences, Media Psychology Lab, KU Leuven , Leuven,; Research Foundation Flanders (FWO-Vlaanderen) , Brussels,

Abstract. Abstract Amid growing concerns about smartphones and students' mental health, schools worldwide are introducing smartphone policies, often without evidence on their effectiveness. This longitudinal study examines how three naturally implemented policies (full restrictions, partial restrictions allowing use during lunchbreak, partial restrictions allowing use during lunch, and afternoon break) relate to mental health across one school year (up to seven waves). Differences were observed at the first measurement wave (two months after implementation): Students in the full-restriction school reported highest depressive symptoms and anxiety. Differences in life satisfaction were also observed without a clear pattern across policy types. Growth trajectory analyses revealed no significant differences in mental health changes over time. Yet, within schools, both the full-restrictions and the lunchbreak school showed increases in depressive symptoms. Life satisfaction decreased slightly in the lunchbreak school. These findings suggest that smartphone restrictions are not a simple solution to youth mental health challenges.

Read article →

Nature

Educational intervention can rewire village social networks

2026-10-14 · DOI 10.1038/S41586-026-11154-7

Marios Papamichalis; Laura Forastiere; Edoardo M. Airoldi; Nicholas A. Christakis

Abstract. Social networks shape behaviour, affect information dissemination and undergird collective action. And they can be exploited in the design of interventions in these domains1-7. Although social networks obey fundamental principles, such as the degree distribution8,9, they can nevertheless evolve as ties form and break. Here, through a randomized controlled trial involving a total of 8,331 people in 110 remote Honduran villages, we evaluate the secondary effects of a 22-month public-health intervention1 on pre-existing social-network structure. We used a design in which a varying fraction of village households received the intervention, and villages in which no one received the intervention were used as the primary reference group. In 'low-dosage' villages (in which 5-30% of households received the intervention), people who received the intervention tended to sever ties with untreated individuals to whom they previously went for health advice. Conversely, in high-dosage villages (50-100%), treated individuals increased their ties. These average changes are explained mainly by tie dynamics; that is, treated-treated pairs persist and form new links, whereas treated-untreated ties dissolve. A mediation analysis indicates that absorption of health information increases the likelihood of tie formation and persistence. The rewiring patterns generally lasted for three years after the intervention ceased. Moreover, although the intervention delivered health information, it reshaped friendship and financial ties, not just health-advice connections. In aggregate, the introduction of a new information regime acted as a kind of social institution, rewiring individuals' connections even though the overall degree distribution of village-wide networks remained relatively stable.

Read article →

Nature Communications

Current aviation roadmaps are not within planetary boundaries

2026-10-13 · DOI 10.1038/S41467-026-77472-6

Bastien Pais; Alexandre Gondran; Lorie Hamelin; Florian Simatos

Abstract. Abstract Aviation decarbonization roadmaps predominantly rely on replacing fossil kerosene with alternative jet fuels, yet the systemic impact of these alternatives on planetary boundaries other than climate change has not been quantified. Here we integrate prospective life-cycle inventories with the planetary boundaries framework to evaluate the absolute impacts of aviation roadmaps as time-dependent trajectories, rather than endpoint snapshots. Across scenarios to 2050, we find that fuel-switch-centered roadmaps do not achieve sustainability, persistently transgress the climate change and biosphere integrity core boundaries, and exert large demands for energy, land, freshwater and CO 2 capture that challenge their feasibility. Slower traffic growth brings aviation closer to its share of the safe operating space in terms of lowering energy demand, but respecting additional planetary boundaries requires air traffic to decline by at least 7.2% yr 1 . Our results show that aligning aviation with Earth's safe operating space cannot be achieved through fuel substitution alone and requires governance strategies that combine supply-side solutions with explicit demand-side measures.

Read article →

Unmet care needs at the end-of-life among older adults with activity limitations

2026-10-13 · DOI 10.1038/S41467-026-74719-0

Chengxu Long; Zhuoer Lin; Hazal Swearinger; Jennifer Ailshire; Yuting Qian; Wai-Kit Ming; Huitong Zhao; Xiaohui Hou; Bei Wu; Wei Yang; Xi Chen; Shanquan Chen

Abstract. Despite the growing global demand for end-of-life care resulting from rapid social transformations, significant gaps exist in cross-national research examining unmet end-of-life care needs. Using harmonized end-of-life survey data (2010-2021) from the United States, 28 European countries, South Korea, China, and Mexico, here we show that the United States, Europe, and South Korea exhibit higher overall unmet end-of-life care needs than China and Mexico, where limited formal care provision is compensated with high levels of informal care. Death at institutions is associated with lower overall unmet end-of-life care needs in the United States and South Korea, with Mexico presenting an exception. Similarly, urban areas in Europe and South Korea exhibit lower unmet formal end-of-life care needs compared to rural areas. Among married individuals, only Mexico exhibits significantly lower unmet overall end-of-life care needs compared to unmarried individuals. Having children is consistently linked to lower unmet informal end-of-life care needs across the United States, Europe, and South Korea. Our findings highlight the complex interplay between care systems and family demographic structures in shaping end-of-life care access, and underscore the need for tailored policy approaches that strengthen both formal and informal care systems while addressing sociodemographic disparities in care access. Formal end-of-life care expansion in the US, Europe, and South Korea has not fully compensated for declining informal care, resulting in care gaps. Demographic transitions interact with development contexts to create distinct care landscapes.

Read article →

Block-fitness modeling of the global air mobility network

2026-10-12 · DOI 10.1038/S41467-026-78115-6

Giulia Fischetti; Anna Mancini; Giulio Cimini; Jessica T. Davis; Abby O. K. Leung; Alessandro Vespignani; Guido Caldarelli

Abstract. Abstract Accurate representations of origin-destination (OD) passenger flows on the World Air Transportation Network (WAN) are fundamental inputs to models of global mobility, epidemic risk, and infrastructure planning. However, OD data are largely commercial and proprietary, therefore often inaccessible to the research community. Here we introduce a generative model that treats air mobility as a stochastic process within a sparse maximum-entropy framework, informed by aggregate traffic volumes of individual locations and geographic regions. The model accurately reconstructs both network structure and mobility flows, yielding epidemic spreading trajectories that closely match those obtained using real mobility data. Furthermore, the method can be effectively implemented using open data on flight routes and airport traffic, reducing dependence on proprietary datasets. Our approach provides a scalable, interpretable, and computationally efficient framework for scenario analysis and policy design in global mobility systems.

Read article →

Nature Human Behaviour

Simulated responses from large language models are weak proxies for human economic preferences across 74 countries

2026-10-12 · DOI 10.1038/S41562-026-02611-4

Fadong Chen; Qijun Wang; Lei Wang; Sudeep Bhatia

Abstract. Large-scale surveys are central to investigating economic preferences but are costly and slow. Can large language model (LLM)-simulated responses substitute for human survey data? We conducted a virtual Global Preference Survey, simulating six experimentally validated economic preferences for 67,201 matched human respondents across 74 countries. Across 20 LLM configurations with diverse prompts and robustness checks, we found systematic but modest individual-level alignment (87.5% of Spearman correlations have 95% confidence intervals (CIs) above zero), with LLMs recovering the direction of associations in 77.8% (95% CI, (71.6%, 83.1%); CI-based consistency ratio) of gender, age and cognitive-ability dimensions, but only 48.1% (95% CI, (45.1%, 51.2%)) of cultural dimensions. Ablation and error decompositions suggest weaker alignment when removing demographic information than when removing country labels, and in many models alignment is weaker in economically undeveloped and less digitally connected countries. LLM-simulated responses may recover some individual- and group-level patterns but are not yet reliable proxies for cross-national representative surveys in macro-level inference or policy evaluation.

Read article →

Political Analysis

The Synthetic Imputation Approach: Using LLMs to Generate Synthetic Texts for Underrepresented Categories in Supervised Classification Tasks

2026-10-14 · DOI 10.1017/PAN.2026.10057

Joan C. Timoneda

Purdue University

Abstract. Abstract Annotated text data are central to many empirical applications in political science, yet many annotation tasks involve categories that are sparsely populated. These low- N categories pose problems for both encoder-only and decoder-only (generative) Transformer large language models (LLMs), leading to low performance, biased evaluation and measurement error in downstream analyses. This article introduces the synthetic imputation approach, a targeted method for expanding sparse categories while remaining consistent with the category-defining content of the human-coded examples. The approach uses a generative LLM to produce additional texts for underrepresented categories by conditioning on small, rotating subsets of human-coded examples. Rather than paraphrasing or replacing the original data, synthetic imputation adds informative variation only where data are thin. I evaluate the approach using both encoder-only and decoder-only LLM classifiers across multiple political science applications. The results show that synthetic imputation substantially improves performance and evaluation stability in sparse settings, recovering full-sample performance once a modest number of true observations is available, with limited and predictable overfitting at very low levels of N .

Read article →

Proceedings of the National Academy of Sciences

AI can persuade people to take political actions

2026-10-16 · DOI 10.1073/PNAS.2611368123

Kobi Hackenburg; Luke Hewitt; Caroline Wagner; Christopher Summerfield; Ben M. Tappin

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Abstract. There is substantial concern about the ability of advanced AI to influence people's behavior. A rapidly growing body of research has found that AI can produce large persuasive effects on people's attitudes, but whether AI can persuade people to take consequential real-world actions has remained unclear. In two large preregistered experiments (N = 17 , 950 responses from 14,779 people), we used conversational AI models to persuade participants on a range of attitudinal and behavioral outcomes, including signing real petitions and donating money to charity. We found sizable AI persuasion effects on these behavioral outcomes (e.g. + 19.7 percentage points on petition signing). However, we observed no evidence of a correlation between AI persuasion effects on attitudes and effects on behavior. Moreover, we replicated prior findings that information provision drove effects on attitudes, but found no such evidence for our behavioral outcomes. In a test of eight behavioral persuasion strategies, the strategy focused on information provision produced the smallest behavioral effect, though differences among the eight were small. Taken together, these results suggest that previous findings relying on attitudinal outcomes may generalize poorly to behavior, and therefore risk substantially mischaracterizing the real-world behavioral impact of AI persuasion.

Read article →

Insights from coproduced scenarios toward urban sustainability transformations

2026-10-16 · DOI 10.1073/PNAS.2537819123

David M. Iwaniec; Nancy B. Grimm; Marta Berbes Blazquez; Elizabeth M. Cook; Tischa A. Munoz-Erickson

Cary Institute of Ecosystem Studies; Environmental Science Department, Barnard College; International Institute of Tropical Forestry, United States Department of Agriculture Forest Service; School of Environment, Resources, and Sustainability, University of Waterloo; School of Life Sciences, Global Institute of Sustainability and Innovation, and Global Futures Lab, Arizona State University; Urban Studies Institute, Andrew Young School of Policy Studies, Georgia State University

Abstract. Urban sustainability transformations call for fundamental shifts in how cities are organized, developed, and governed. Yet we need a better understanding of the diversity of visions and pathways that can enable transformative change. We address this gap through an analysis of 60 coproduced positive scenario visions and pathways developed using the Sustainable Future Scenario process, across nine US and Latin American cities participating in the Urban Resilience to Extremes Sustainability Research Network. We engaged diverse urban stakeholders-including government, nonprofit, and private decision-makers, community leaders, residents, and academic collaborators-in more than 20 coproduction workshops to explore alternative urban futures of sustainability and resilience. We examined how the coproduced scenarios represent five domains of transformative change (nature-human relationships, economic transformation, governance, social and environmental justice, and infrastructure futures) and four theories of transformative change (structural/institutional, agency/power, goals of the system, and worldviews/values). Most pathways combined multiple theories of change operating across multiple domains, revealing urban transformations as entangled rather than singular interventions. Some strategies were common across cities-car-free regions, decentralized infrastructure, community-based governance, expanded fundamental rights-yet were pursued with different transformative aims, demonstrating how a strategy's transformative intent depends on the theories of change behind it. These findings position coproduced scenarios as more than a normative or planning tool and offer an analytical approach for surfacing the possibility space of urban transformation.

Read article →

Estimating the threat of AI-agent responding across online survey platforms

2026-10-15 · DOI 10.1073/PNAS.2606460123

Grace Zhang; Robert Walatka; Stephanie Y. Chen; Alexander K. Moore; Oleg Urminsky; Kiante Fernandez; Andrea Low; Jonathan E. Bogard; Craig R. Fox

Behavioral Decision Making Area, Behavioral Decision Making Area, Anderson School of Management, University of California; Behavioral Science, University of Chicago Booth School of Business; Department of Psychology, University of California; Marketing, London Business School; Marketing, University of Chicago Booth School of Business; Marketing, University of Illinois Chicago College of Business Administration; Organizational Behavior, Olin Business School, Organizational Behavior, Olin Business School, Washington University in St. Louis

Abstract. Recent research and advances in LLMs have led to widespread concern that AI agents could pose as human online survey-takers. However, it remains unclear how prevalent AI agents are on survey research platforms and how to effectively detect AI agents. We validated a series of AI detection tests that effectively separated verified-human participants from both prompted commercial and purpose-built AI agents. Using these tests, we collected surveys on seven online platforms and find wide variation in rates of respondents failing AI tests, ranging from 6 to 41% across platforms (compared to a 2.4% false-positive rate for in-person human participants). We further demonstrate that undetected AI agents can impact the results of online surveys. Our findings suggest that while there is an urgent need for AI detection tests and consistent, systematic monitoring of data quality on online platforms, some platforms currently appear to provide data with a low rate of AI agents.

Read article →

Suspected distortion of citations in high-impact cancer journals

2026-10-15 · DOI 10.1073/PNAS.2618402123

Baptiste Scancar; Jennifer A. Byrne; David Causeur; Adrian G. Barnett

Department of Statistics and Computer Science, Institut de Recherche Mathematique de Rennes UMR 6625 CNRS, L'Institut Agro; New South Wales Health Statewide Biobank, New South Wales Health Pathology; School of Medical Sciences, Faculty of Medicine and Health, The University of Sydney; School of Public Health and Social Work, Queensland University of Technology

Abstract. Research and scholarship are shaped by article citations, which underpin the communication of ideas, assignment of credit, journal impact factors, and author career progression. Given their key influence on author and journal metrics, citations can be intentionally manipulated to inflate the reputation of journals and researchers. Paper mills, unethical organizations that produce and sell manuscripts and publishing services, may also be manipulating citations, but the extent of this manipulation is unknown. Here, we show that molecular cancer articles sharing features with retracted papers from paper mills display citation patterns that suggest systematic inflation. These articles were published in journals in the top decile of journal rankings. Suspected paper mill articles received 50 to 100% more citations than other papers 1 to 3 y after publication, while paradoxically attracting fewer readers and online accesses. Suspected paper mill articles also cited-and were cited by-other suspected paper mill articles, which frequently appeared in journals previously reported as paper mill targets. Citations from suspected paper mill articles measurably inflated journal citation metrics. These findings suggest that paper mills inflate the citation metrics of supported publications and affected journals at scale. This widespread manipulation of citation metrics may amplify unreliable findings, slowing scientific progress and spurring unreasonable citation benchmarks for research articles, journals, and authors. Our findings point to measurable signatures of citation manipulation that could support article- and journal-level screening.

Read article →

AI-generated language remains systematically distinguishable from human communication

2026-10-12 · DOI 10.1073/PNAS.2531748123

Nicolo Pagan; Petter Tornberg; Christopher A. Bail; Christopher Barrie

Department of Informatics, University of Zurich; Department of Sociology, Computer Science, Political Science, and Public Policy, Duke University; Department of Sociology, New York University; Department of Sociology, University of Oxford; Institute for Logic, Language and Computation, University of Amsterdam

Abstract. Large language models (LLMs) are increasingly used in the social sciences to simulate human behavior and communication, yet validation methods for these applications remain limited. Existing evaluations often rely on human judgments of plausibility, a coarse, difficult-to-scale, and often unreliable benchmark for distinguishing AI-generated from human text. As a result, researchers lack robust tools for assessing the realism of LLM-generated language and for calibrating models to real-world data. We address this gap in two ways. First, we introduce the Conversational Human Authenticity Test (CHAT), a validation framework that combines aggregate metrics of detectability and semantic similarity with interpretable linguistic features, including stylistic markers and topical patterns, to assess how closely LLMs approximate human language in a given dataset. Second, we benchmark nine open-weight LLMs across five calibration strategies, including fine-tuning, stylistic prompting, and context retrieval, on user interactions from X, Bluesky, and Reddit. We find that even after calibration, LLM outputs remain clearly distinguishable from human text, especially in affective tone and emotional expression. Instruction-tuned models generally underperform base models, and larger models do not obviously improve human-likeness. We also identify a trade-off between realism and semantic fidelity: methods that make outputs seem more human often move them further from the content of human responses. Together, these results provide a scalable framework for validation and calibration in LLM-based social simulation while underscoring current limits in reproducing human communication.

Read article →

Integrating cooperative breeding with reciprocity and culture extends Hamilton's rule to humans

2026-10-12 · DOI 10.1073/PNAS.2617040123

Alessandra Cassar; Daniel Friedman; Shreeja Guha; Timothy Wilkins

Department of Economics, University of California; Department of Economics, University of Essex; Department of Economics, University of San Francisco

Abstract. We study how kinship and reciprocity shape giving and child care support in societies with limited access to markets. Extending Hamilton's rule to incorporate reciprocity, we test it using behavioral data from dictator games and surveys in remote matrilocal and patrilocal villages of northeast India (N = 406 parents of young children). We replicate with data from urban Manila neighborhoods (N = 301) and the Solomon Islands (N = 624). Consistent with predictions, transfers to recipients increase with genetic relatedness to the donor and even more with relatedness to the donor's child-a hallmark of reciprocity anchored in shared genetic investment. Nonanonymous conditions amplify giving, especially to nonkin. Cultural norms shape cooperation, but do not trump genetic relatedness. These results underscore the role of affinal ties and reciprocal exchange in sustaining cooperation, and suggest a mechanism through which cooperative breeding could contribute to the emergence of generalized reciprocity and human sociality.

Read article →

Laundering and evasion in response to antislavery policies in Brazil's cattle sector

2026-10-12 · DOI 10.1073/PNAS.2532121123

Tara Mittelberg; Marin E. Skidmore; Juliana L. S. C. Brandao; Lisa L. Rausch; Holly K. Gibbs

Department of Agricultural and Applied Economics, University of Wisconsin-Madison; Department of Agricultural and Consumer Economics, University of Illinois at Urbana-Champaign; Nelson Institute for Environmental Studies, Center for Sustainability and the Global Environment, University of Wisconsin-Madison

Abstract. Modern slavery remains pervasive across many economic sectors. Progress toward eradicating it has been partially hindered by a lack of tested, comprehensive policy responses. Using data on millions of cattle transactions in Brazil, we evaluated how ranchers changed their selling behavior after being added to the "Dirty List," a public registry that names and shames employers found to use modern slavery. Dirty Listed ranchers significantly reduced direct sales to slaughterhouses that signed sustainable supply chain commitments but evaded sanctions by laundering cattle through compliant intermediaries or selling to nonsignatory slaughterhouses. These responses weakened the financial cost of listing to ranchers, and likely allowed beef and leather linked to modern slave labor to enter international supply chains. Our findings highlight the limits of downstream enforcement and underscore the need to monitor upstream suppliers to ensure slavery-free supply chains.

Read article →

Multidimensional vulnerabilities of informal settlements in the Urban South

2026-10-12 · DOI 10.1073/PNAS.2523034123

Mingxing Chen; Wenming Song; Dadao Lu; Yu Deng; Chenglong Wang; Huiping Jiang; Yuyu Zhou; Murray W. Scown; Liangkan Chen; Jing Ma; Heran Zheng; Xuemei Bai

College of Resource and Environment, University of Chinese Academy of Sciences; Department of Geography and Institute for Climate and Carbon Neutrality, The University of Hong Kong; Fenner School of Environment and Society, Australian National University; Institute for Future Initiatives, The University of Tokyo, 7-3-1 Hongo; Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences; Key Laboratory of Regional Sustainable Development Modeling, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences; Lund University Centre for Sustainability Studies, Lund University; School of Geography and Ocean Science, Nanjing University; The Bartlett School of Sustainable Construction, University College London

Abstract. Informal settlements are widespread and rapidly growing, often associated with high vulnerabilities. However, global-scale understanding on their spatial distribution and complexity of vulnerability is limited. Here, we delineate the informal settlements across a sample of 1,914 cities of the Global South by integrating Sentinel-2 imagery with deep learning models, and we explore the heterogeneity of vulnerability using Urban Informal Settlement Vulnerability Index that encompasses living conditions, social infrastructure, and environmental risk. We found that Urban Informal Settlements cover 12.94 2.52% of their total urban areas (approximately 34.04 10 3 km 2), especially in sub-Saharan Africa, South Asia, and Middle Asia. Sub-Saharan African cities have the highest vulnerability of informal settlements, with 31.75 6.20% of total urban areas being occupied by informal settlements, while East Asian cities exhibit relatively low vulnerability across all dimensions. We also find that urban informal settlements with high environmental exposure are widespread in the Global South, and there is a significantly polarized distribution between "low" and "high" vulnerability categories. The findings highlight the disparate distribution of multidimensional vulnerability in informal settlements across the Urban South, providing a comparative evidence base for further local assessment and context-specific policy discussions related to SDG 11.

Read article →

Science

Market dilution, fair use, and generative AI

2026-10-15 · DOI 10.1126/SCIENCE.AEL4888

Pamela Samuelson; Abhishek Nagaraj; Christopher Jon Sprigman

Haas School of Business, University of California, Berkeley, CA, USA.; School of Law, New York University, New York, NY, USA.; School of Law, University of California, Berkeley, CA, USA.

Abstract. Market dilution theory, which could affect AI use in science, requires conceptual clarity and empirical evidence

Read article →

Science Advances

Global conservation additionality hindered by persisting fishing pressure in marine protected areas

2026-10-16 · DOI 10.1126/SCIADV.AED6878

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Abstract. Marine protected areas (MPAs) are the cornerstone of global marine conservation strategies. Yet, their capacity to achieve conservation gains (i.e., conservation additionality) hinges on the decrease in human threats to biodiversity within their boundary, a feature rarely measured or reported. Here, we test a central but underexamined question in global conservation policy: Have MPA implementations actually reduced within-boundary fishing pressure, currently the main direct human threat to marine biodiversity? Using a stratified before-after-control-impact approach, we investigated 861 MPAs of varying levels of protection across all ocean basins, representing half of global MPA coverage, and found that most implementations were not followed by decreases in industrial fishing effort. Protection levels strongly predicted the direction and magnitude of relative changes in fishing effort after implementation. Only fully protected areas were associated with significant decreases in fishing effort within boundaries. Highly protected areas were only associated with lower rates of increase in fishing effort within compared to outside boundaries. Both fully and highly protected areas were implemented in areas experiencing initial low fishing effort, reducing the overall abatement in fishing effort achieved. Conversely, areas with higher initial fishing effort tended to be attributed to MPAs with low levels of protection, also limiting abatement. Larger MPAs were associated with greater decreases in fishing effort, while time since implementation had no significant effect. Complementing targets of protection coverage and protection levels with conservation additionality targets, such as pressure abatement, could shift incentives from quantity to effectiveness of interventions and ensure positive biodiversity conservation outcomes.

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How inequality became durable before states: Evidence from 6000 years in the Central Plain of China

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Abstract. How did durable inequality emerge before the rise of states? Drawing on a synthesis of previously published mortuary data from more than 2300 burials, together with isotopic, zooarchaeological, and osteological data spanning nearly 6000 years (7000 to 1500 BCE) in the prehistoric Central Plain of China, we reconstruct long-term patterns of relational mortuary inequality. We show that inequality followed a strongly nonlinear trajectory and cannot be explained primarily by agricultural intensification or population scale. Instead, substantial disparities were structured by differential positions within social-relational networks that mediated access to labor, resources, and symbolic capital. Transformations in subsistence, the rise of mobile and appropriable wealth, and escalating conflict reorganized these networks, concentrating ritual and political-military power and technological control. Early states stabilized rather than created these preexisting disparities by institutionalizing control over resources, violence, and network-based access. These findings highlight the central role of social networks in shaping durable inequality before formal state institutions.

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The network structure of the urban revolution

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Abstract. Long-distance interaction has deep prehistoric roots, but the networks that emerged during the Urban Revolution were unprecedented in scale and organization. How early cities sustained this extensive connectivity in the absence of formal markets or centralized states remains a fundamental question. Focusing on the Uruk Expansion in Mesopotamia (ca. 3900 to 3000 BCE), we assemble the largest spatial dataset currently available and reconstruct transportation networks across three successive phases. Using network analysis, we show that an initially dendritic, river-based system evolved into modular regional clusters and ultimately into a more integrated structure linked by selective long-distance shortcuts. This transformation reduced dependence on key intermediaries and generated small world-like patterns combining strong local cohesion with efficient system-wide reach. These results demonstrate that decentralized interaction processes, supported in part by mobile intermediary communities, played a central role in sustaining early urban connectivity.

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Previous Week Catch-Up

Communication Research

Human Imagery and Psychological Reactance in Persuasive Health Messages

2026-10-11 · DOI 10.1177/00936502261478982

Chelsea L. Ratcliff; Helen Lillie; Manusheela Pokharel; Jakob D. Jensen; Doris E. Acheme; Benjamin Sam-Paintsil

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Abstract. Although mass-mediated persuasive messages often feature images of people, their impacts are surprisingly understudied. We tested whether attributes of human imagery influence psychological reactance to a health promotion PSA, either alone or combined with language style. Participants (N = 1,100) viewed a message promoting sun safety that contained a forceful (vs. courteous) appeal and featured a model with forceful (vs. neutral) body language. The model's physical characteristics (gender and skin tone) also varied. Both forceful text and forceful body language indirectly lowered protective behavioral intentions via reactance, and male-model messages had this effect for female participants. Our findings demonstrate the potential for features of human imagery to inadvertently hinder persuasion. Study results also highlight the value in measuring perceived visual freedom threat alongside traditional reactance measures. We discuss implications for reactance theory and measurement and highlight important directions for research on reactance to visuals.

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Nature Communications

A dual-attribution framework reveals how foreign investment and trade redistribute cobalt availability

2026-10-10 · DOI 10.1038/S41467-026-78222-4

Lei Liu; Ling Zhang; Songyan Jiang; Yu Li; Ruishi Li; Zhu Mei; Zengwei Yuan; Jun Chen

Abstract. Global resource allocation is increasingly shaped by foreign direct investment (FDI) and international trade, yet their combined effects on distribution of critical mineral availability across countries remain poorly quantified. Here, we develop a dual-attribution framework covering 193 countries and 36 territories that attributes primary cobalt availability according to corporate ownership and secondary availability according to international trade. We compare these estimates with a combined baseline scenario using geographic-production and pre-trade perspectives to assess how ownership and trade reshape country-level cobalt availability. Relative to this baseline, total cobalt availability is higher in 63.6% of countries, while 72.7% show a positive composite change when primary and secondary availability are equally weighted. Although these mechanisms do not alter the global total amount of cobalt available, they reduce its concentration across countries. Our findings highlight how FDI and trade reshape cobalt distribution and support resource-governance policies that account for both host and exporting countries. This study shows how foreign investment and international trade reshape global cobalt availability. A dual-attribution framework reveals higher attributed availability in most countries while reducing country-level concentration.

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Political Communication

Who Verifies the Verifiers? AI, Local Knowledge Production, and Credibility in Tanzania's 2025 Election

2026-10-11 · DOI 10.1080/10584609.2026.2745832

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Abstract. This study examines how Factchecking AI tools and community-based credibility practices interacted during Tanzania's 2025 presidential election. Using data from five regions, we analyzed 342 media materials, 87 interviews, field observations, and 22 distinct contested election events. The findings identify three forms of verification disjuncture. First, AI-assisted translation and transcription sometimes changed the meaning of politically important statements before journalists could verify them. Second, fact-checks and disputed claims often moved through different communication channels, which limited the chances that people would see the correction. Third, some communities trusted personal experience, familiar sources, and collective discussion more than professional fact-checkers. Participants judged credibility through relationships, lived experience, group discussion, humor and satire, and social accountability. These practices sometimes supported the same conclusions as institutional verification and sometimes produced different judgments. We argue that fact-checking becomes politically effective when the meaning of a claim is preserved, the correction reaches the people discussing it, and the source of the correction is recognized as credible. This explains why an accurate fact-check can still have limited public influence.

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